	AIR UNIVERSITY
	DEPARTMENT OF COMPUTER ENGINEERING
	EXPERIMENT NO: 3

Lab Tittle: Cisco Router User Interface

Name:	Muhammad Waqas	**Reg No:**	240756
	Faiq Ali		249520

Objectives: To familiarize students with routers and how to connect a router for configuration.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks:_____ Obtained Mark: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____



DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Introduction to Hub, Switch & Router

Lab Report: 3

Tittle: Cisco Router User Interface

Name: Muhammad Waqas, Faiq Ali

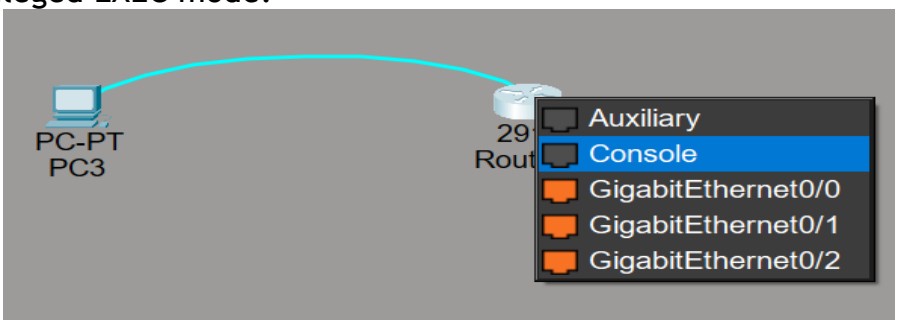
Reg. No: 240756, 249520

Submitted To: CE- Engr. Sir Basharat Ahmad

Date: 29 Sep, 2026

Lab Task 1:

Connect a Cisco router via the console port and access the command line. Verify that you can enter privileged EXEC mode.



User to Privileged EXEC mode:

```
Router>enable
Router#
```

Basic Configuration Setup

Setup Mode: A step-by-step guide to configuring the router if no initial configuration is present

Physical | Config | Desktop | Programming | Attributes

Terminal

```
Router>
Router>enable
Router#setup


         --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]:yes

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.


Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: yes
Configuring global parameters:

  Enter host name [Router]: My-Router

  The enable secret is a password used to protect access to
  privileged EXEC and configuration modes. This password, after
  entered, becomes encrypted in the configuration.
  Enter enable secret: waqas123
```

```
  The enable password is used when you do not specify an
  enable secret password, with some older software versions, and
  some boot images.
  Enter enable password: waqas321

  The virtual terminal password is used to protect
  access to the router over a network interface.
  Enter virtual terminal password: 1234
Configure SNMP Network Management? [no]:no

Current interface summary

Interface              IP-Address      OK? Method Status
Protocol

GigabitEthernet0/0     unassigned      YES manual administratively
down down

GigabitEthernet0/1     unassigned      YES manual administratively
down down

GigabitEthernet0/2     unassigned      YES manual administratively
down down

Enter interface name used to connect to the
management network from the above interface summary:
GigabitEthernet0/0

Configuring interface GigabitEthernet0/0:
  Configure IP on this interface? [yes]: yes
    IP address for this interface: 192.168.1.1
    Subnet mask for this interface [255.255.255.0] : 255.255.255.0

The following configuration command script was created:

!
hostname a
enable secret 5 $1$mERr$Tg1seMrv6KpaRO3xYq1uo.
enable password as
line vty 0 4
password asd
!
interface Vlan1
 shutdown
 no ip address
!
interface GigabitEthernet0/0
 no shutdown
```

```
 ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet0/1
 shutdown
 no ip address
!
interface GigabitEthernet0/2
 shutdown
 no ip address
!
end

[0] Go to the IOS command prompt without saving this config.
[1] Return back to the setup without saving this config.
[2] Save this configuration to nvram and exit.

Enter your selection [2]: 2
Building configuration...
[OK]
Use the enabled mode 'configure' command to modify this
configuration.
```

Extended Setup:

Allows for a more detailed configuration, including hostname, IP addresses, and passwords.

Terminal

```
My-Router>
My-Router>enable
Password:
My-Router#setup


         --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]:yes

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.


Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: no
First, would you like to see the current interface summary? [yes]:no
Configuring global parameters:

  Enter host name [My-Router]: My-Router
```

```
  The enable secret is a password used to protect access to
  privileged EXEC and configuration modes. This password, after
  entered, becomes encrypted in the configuration.
  Enter enable secret: waqas123

  The enable password is used when you do not specify an
  enable secret password, with some older software versions, and
  some boot images.
  Enter enable password: waqas321

  The virtual terminal password is used to protect
  access to the router over a network interface.
  Enter virtual terminal password: 1234
Configure SNMP Network Management? [no]:no
Configuring interface parameters:

Do you want to configure Vlan1  interface? [no]:no
Do you want to configure GigabitEthernet0/0  interface? [no]:yes
    IP address for this interface: 192.168.1.1
    Subnet mask for this interface [255.255.255.0] : 255.255.255.0
Do you want to configure GigabitEthernet0/1  interface? [no]:yes
    IP address for this interface: 192.168.1.2
    Subnet mask for this interface [255.255.255.0] : 255.255.255.0
Do you want to configure GigabitEthernet0/2  interface? [no]:yes
    IP address for this interface: 192.168.1.3
    Subnet mask for this interface [255.255.255.0] : 255.255.255.0

The following configuration command script was created:

!
hostname My-Router
enable secret 5 $1$mERr$gnLeLptz5otMNX.XseSe10
enable password waqas321
line vty 0 4
password 1234
!
interface Vlan1
 shutdown
 no ip address
!
interface GigabitEthernet0/0
 no shutdown
 ip address 192.168.1.1 255.255.255.0
!
```

```
interface GigabitEthernet0/1
 no shutdown
 ip address 192.168.1.2 255.255.255.0
!
interface GigabitEthernet0/2
 no shutdown
 ip address 192.168.1.3 255.255.255.0
!
end

[0] Go to the IOS command prompt without saving this config.
[1] Return back to the setup without saving this config.
[2] Save this configuration to nvram and exit.

Enter your selection [2]: 2
Building configuration...
[OK]
Use the enabled mode 'configure' command to modify this
configuration.

My-Router#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

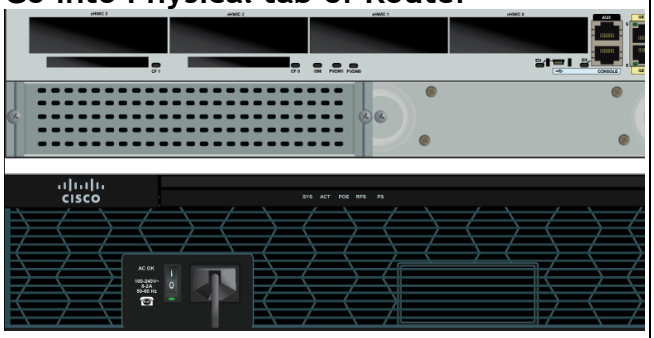
%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down

%SYS-5-CONFIG_I: Configured from console by console
```

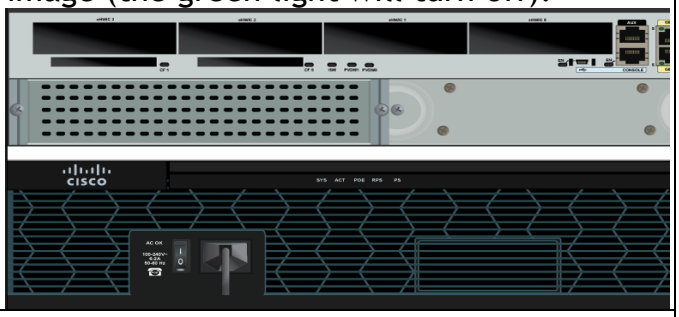
Lab Task 2:

Perform the basic configuration of a Cisco router. Set the hostname, enable secret, and IP addresses for FastEthernet interfaces., also add fastethernet port in router.

Go into Physical tab of Router

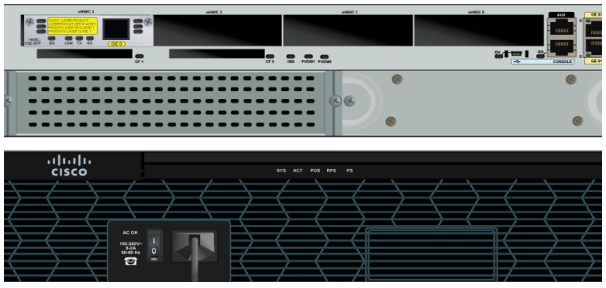


Turn **OFF** the power switch on the router image (the green light will turn off).

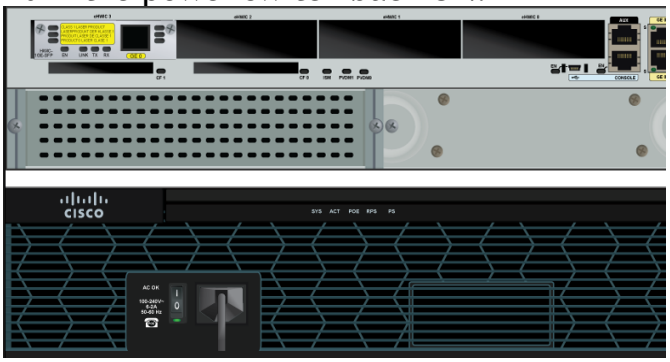


Drag and drop the module into an empty slot on the right side of the router image

HWIC-1GE-SFP



Turn the power switch back **ON**.



```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router-a
Router-a(config)#enable password waqas123
Router-a(config)#enable secret waqas321
Router-a(config)#interface GigabitEthernet 0/0
Router-a(config-if)#ip address 192.168.10.1 255.255.255.0
Router-a(config-if)#no shutdown

Router-a(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router-a(config-if)#exit
Router-a(config)#end
Router-a#
%SYS-5-CONFIG_I: Configured from console by console

Router-a#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Lab Task 3:

Boot a Cisco router and observe the output during the POST. Identify the IOS version and memory details.

Show Version

```
Router-a#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 6 minutes, 10 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
 --More--
```

Boot a Cisco router and observe the output during the POST.

```
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x3bcd3d8
Self decompressing the image :
################################################################################ [OK]
Smart Init is enabled
smart init is sizing iomem
                    TYPE       MEMORY_REQ
     Onboard devices &
           buffer pools       0x022F6000
------------------------------------------------
                  TOTAL:      0x022F6000
Rounded IOMEM up to: 36Mb.
Using 6 percent iomem. [36Mb/512Mb]

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Use, duplication, or disclosure by the Government is
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(c) of the Commercial Computer Software - Restricted
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(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.

            cisco Systems, Inc.
            170 West Tasman Drive
            San Jose, California 95134-1706
```

Lab Task 4:

Configure the router using the Extended Setup option

. Assign IP addresses to both Fast Ethernet interfaces

. Logging into the Router To configure the router:

1. User EXEC Mode: Use this mode to view statistics. o Prompt:
2. Router> 2. Privileged EXEC Mode:

```
          --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#
```

3. Use this mode to view and change the router configuration. o Access by typing enable. o Prompt: Router#

```
Router>
Router>
Router>enable
Router#setup


         --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]:yes

At any point you may enter a question mark '?' for help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.


Basic management setup configures only enough connectivity
for management of the system, extended setup will ask you
to configure each interface on the system

Would you like to enter basic management setup? [yes/no]: no
First, would you like to see the current interface summary? [yes]:yes

Current interface summary

Interface              IP-Address      OK? Method Status                Protocol

GigabitEthernet0/0     unassigned      YES manual administratively down down

GigabitEthernet0/1     unassigned      YES manual administratively down down

GigabitEthernet0/2     unassigned      YES manual administratively down down

Vlan1                  unassigned      YES manual administratively down down

Configuring global parameters:
```

```
  Enter host name [Router]: Router-A

  The enable secret is a password used to protect access to
  privileged EXEC and configuration modes. This password, after
  entered, becomes encrypted in the configuration.
  Enter enable secret: 12345

  The enable password is used when you do not specify an
  enable secret password, with some older software versions, and
  some boot images.
  Enter enable password: 54321

  The virtual terminal password is used to protect
  access to the router over a network interface.
  Enter virtual terminal password: 1111
Configure SNMP Network Management? [no]:no
Configuring interface parameters:

Do you want to configure Vlan1  interface? [no]:no
Do you want to configure GigabitEthernet0/0  interface? [no]:yes
    IP address for this interface: 192.168.1.1
    Subnet mask for this interface [255.255.255.0] : 255.255.255.0
Do you want to configure GigabitEthernet0/1  interface? [no]:yes
    IP address for this interface: 192.168.2.1
    Subnet mask for this interface [255.255.255.0] : 255.255.255.0
Do you want to configure GigabitEthernet0/2  interface? [no]:no

The following configuration command script was created:

!
hostname Router-A
enable secret 5 $1$mERr$/Q/mbs3O9oHsKR7rNG4e81
enable password 54321
line vty 0 4
```

```
password 1111
!
interface Vlan1
 shutdown
 no ip address
!
interface GigabitEthernet0/0
 no shutdown
 ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet0/1
 no shutdown
 ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/2
 shutdown
 no ip address
!
end

[0] Go to the IOS command prompt without saving this config.
[1] Return back to the setup without saving this config.
[2] Save this configuration to nvram and exit.

Enter your selection [2]: 2
Building configuration...
[OK]
Use the enabled mode 'configure' command to modify this configuration.

Router-A#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
```

Lab Task 5:

Terminal

```
Router-A#show running-config
Building configuration...

Current configuration : 807 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router-A
!
!
!
enable secret 5 $1$mERr$/Q/mbs3O9oHsKR7rNG4e81
enable password 54321
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX15245X6Y-
!
line con 0
!
line aux 0
!
line vty 0 4
 password 1111
 login
!
!
!
end
```

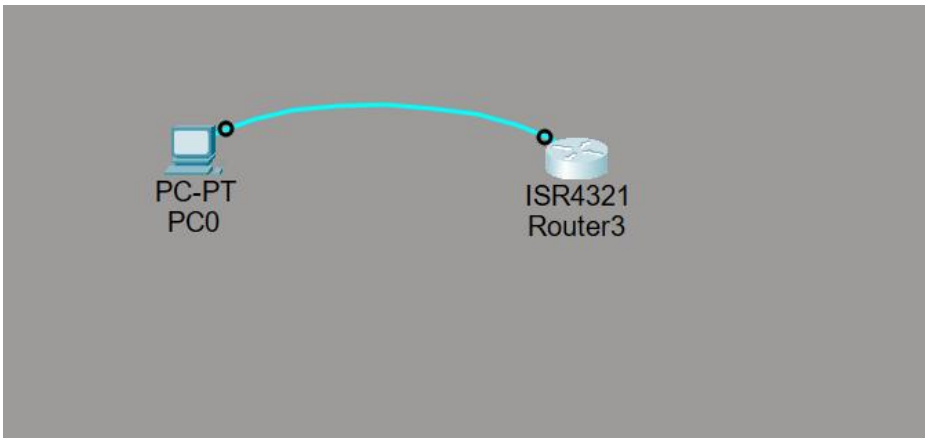
Lab Task 6:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router-b
Router-b(config)#banner motd #
Enter TEXT message.  End with the character '#'.
Warning!Unauthorized access is prohibites #

Router-b(config)#line console 0
Router-b(config-line)#password 12345
Router-b(config-line)#login
Router-b(config-line)#exit
Router-b(config)#enable password 12345
Router-b(config)#enable secret 54321
Router-b(config)#line vty 0 4
Router-b(config-line)#password telnet123
Router-b(config-line)#login
Router-b(config-line)#exit
Router-b(config)#end
Router-b#
%SYS-5-CONFIG_I: Configured from console by console

Router-b#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router-b#
```

Lab task 7:




```
Press RETURN to get started!

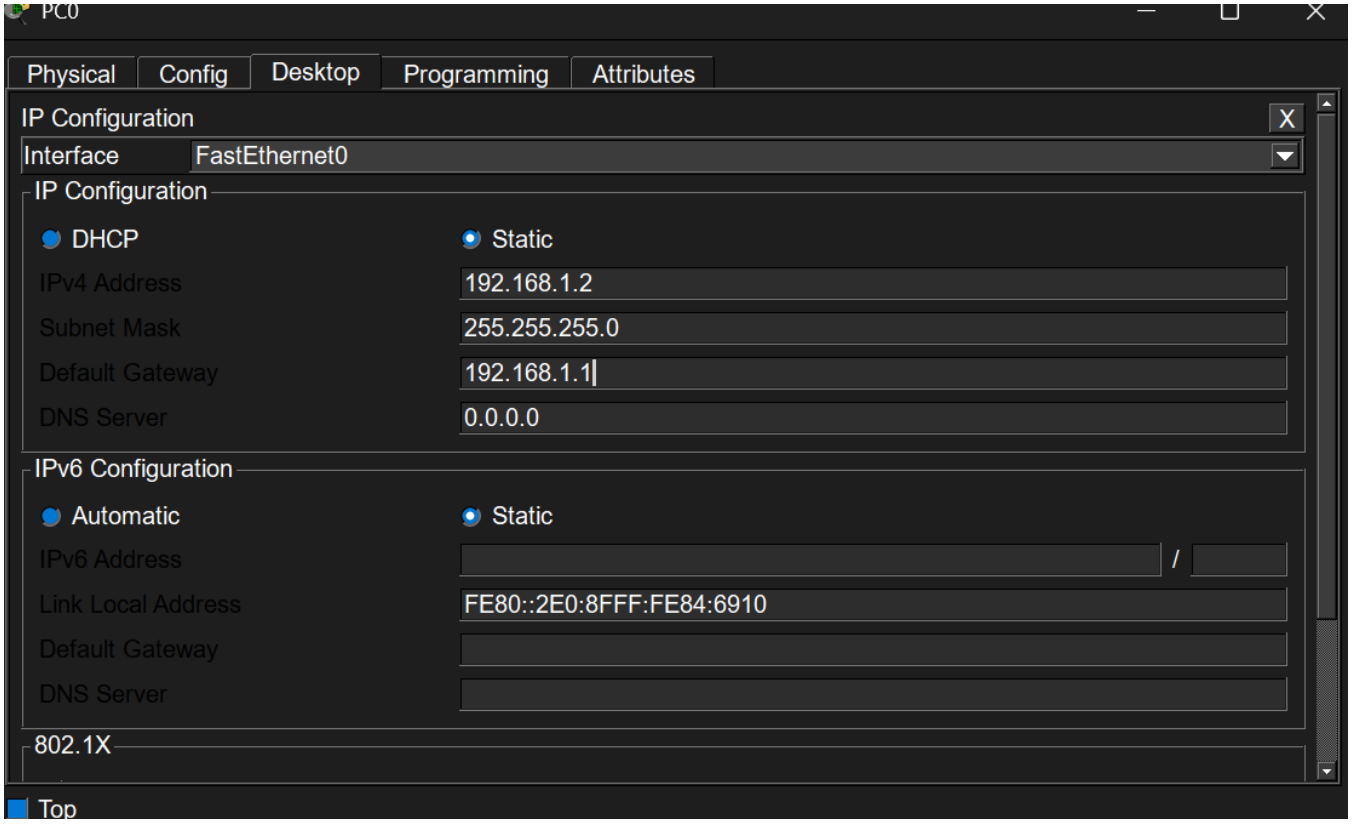


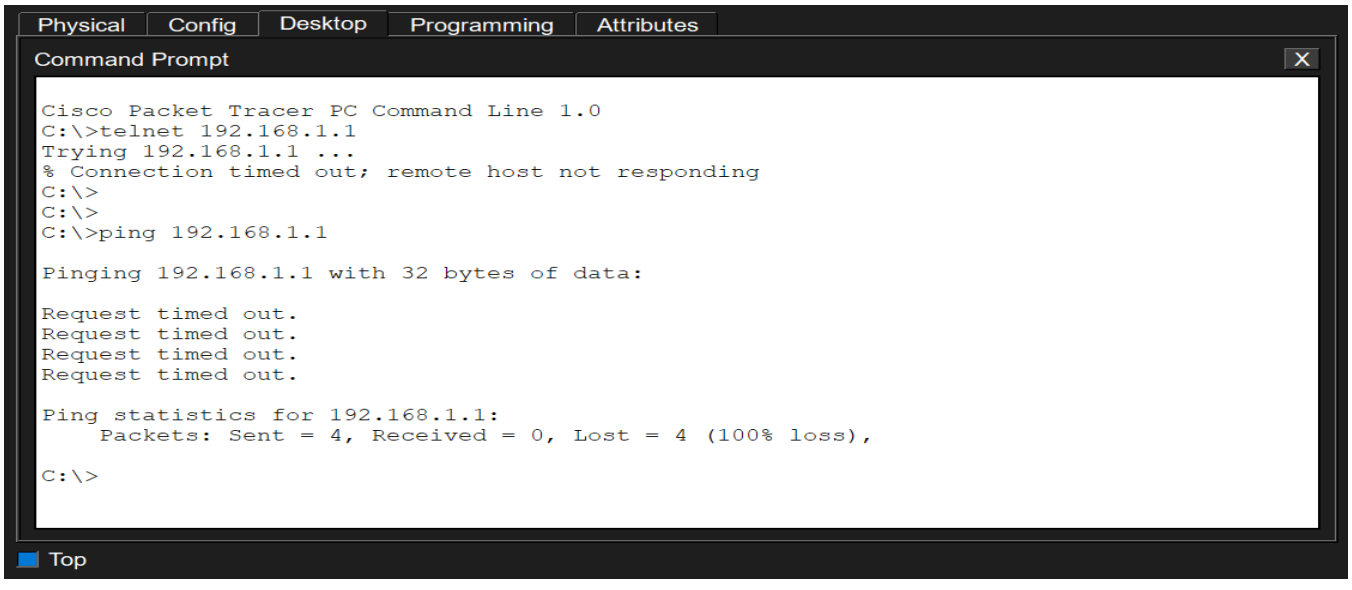
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#line console 0
Router(config-line)#enable secret 1234
Router(config)#line vty 0 4
Router(config-line)#password 32111
Router(config-line)#login
Router(config-line)#exit
Router(config)#interface gigabitEthernet 0/0
%Invalid interface type and number
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#
```


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:8FFF:FE84:6910
Default Gateway
DNS Server
802.1X
Top


Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...
% Connection timed out; remote host not responding
C:\>
C:\>
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router-6
Router-6(config)#enable password 1234
Router-6(config)#enable secret 12345
Router-6(config)#line vty 0 4
Router-6(config-line)#password cisco1234
Router-6(config-line)#login
Router-6(config-line)#exit
Router-6(config)#interface gigabitEthernet 0/0
Router-6(config-if)#ip address 192.168.1.1 255.255.255.0
Router-6(config-if)#no shutdown

Router-6(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router-6(config-if)#exit
Router-6(config)#
```

Post Lab Tasks

Lab task1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router#show version
Cisco IOS XE Software, Version 03.16.05.S - Extended Support Release
Cisco IOS Software, ISR Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version Version 15.5 (3)S5, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2017 by Cisco Systems, Inc.
Compiled Thu 19-Jan-17 11:24 by mcpre

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GPL code under the terms of GPL Version 2.0.  For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
or the applicable URL provided on the flyer accompanying the IOS-XE
software.


ROM: IOS-XE ROMMON

Router uptime is 18 minutes, 23 seconds
Uptime for this control processor is 18 minutes, 23 seconds
System returned to ROM by power-on
System image file is "bootflash:/isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin"
Last reload reason: PowerOn

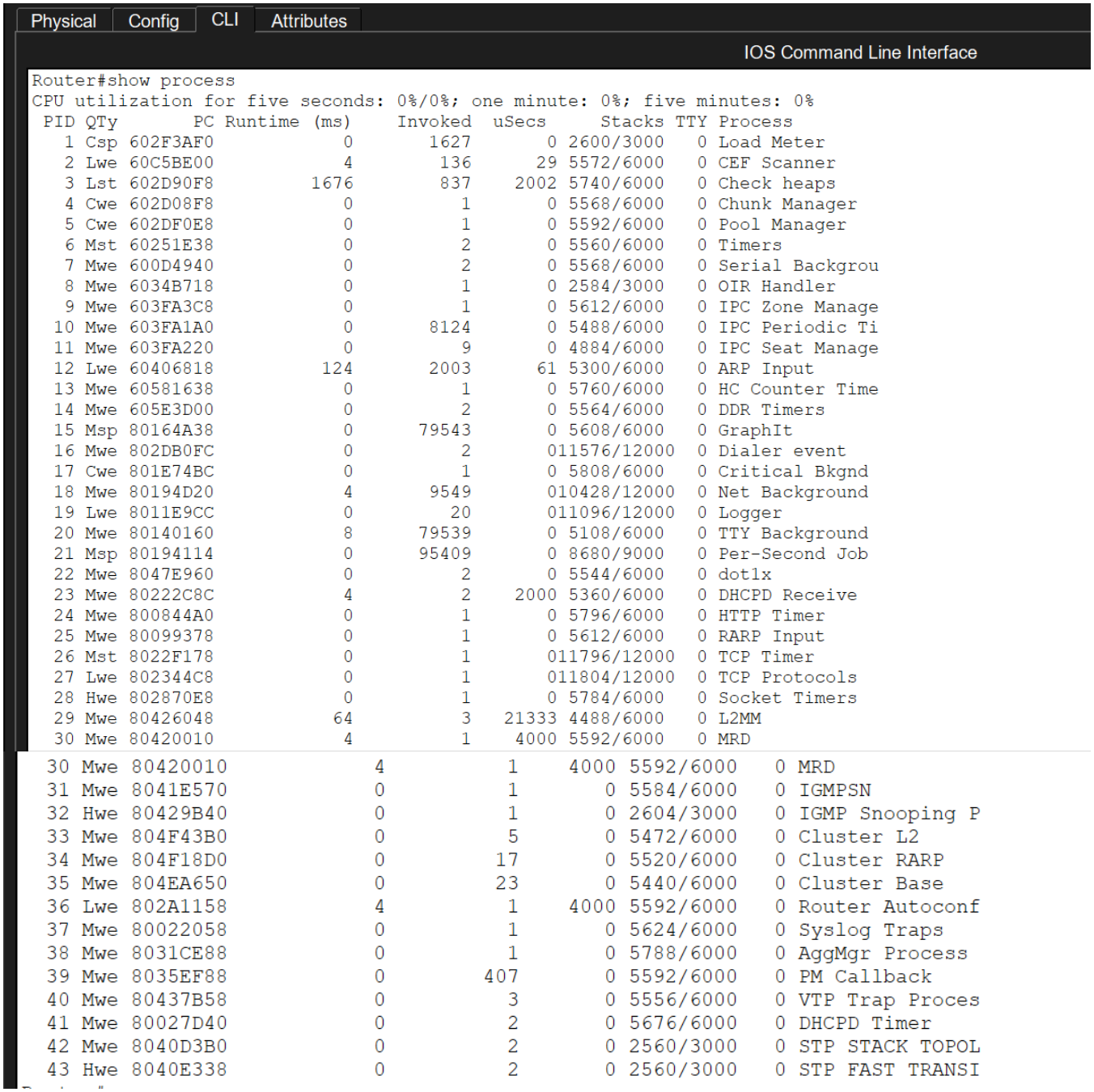

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
```

IOS version: 03.16.05.S version.
Router Model: 2901/k9
RAM: 491520KB / 420 MB
Flash Memory: 32768KB / 32 MB
Router Up Time: 18 minutes 23 sec

Task 2: Viewing IOS Processes

- Objective: Learn how Cisco IOS manages system resources and processes.
- Instructions:

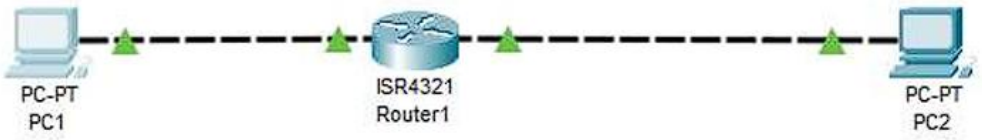
1. Use the command show processes to view the processes running on the router.
2. Identify the most CPU and memory-consuming processes.




```
Router#show process
CPU utilization for five seconds: 0%/0%; one minute: 0%; five minutes: 0%
 PID QTy       PC Runtime (ms)    Invoked  uSecs     Stacks TTY Process
   1 Csp 602F3AF0           0       1627       0 2600/3000   0 Load Meter
   2 Lwe 60C5BE00           4        136      29 5572/6000   0 CEF Scanner
   3 Lst 602D90F8        1676        837    2002 5740/6000   0 Check heaps
   4 Cwe 602D08F8           0          1       0 5568/6000   0 Chunk Manager
   5 Cwe 602DF0E8           0          1       0 5592/6000   0 Pool Manager
   6 Mst 60251E38           0          2       0 5560/6000   0 Timers
   7 Mwe 600D4940           0          2       0 5568/6000   0 Serial Backgrou
   8 Mwe 6034B718           0          1       0 2584/3000   0 OIR Handler
   9 Mwe 603FA3C8           0          1       0 5612/6000   0 IPC Zone Manage
  10 Mwe 603FA1A0           0       8124       0 5488/6000   0 IPC Periodic Ti
  11 Mwe 603FA220           0          9       0 4884/6000   0 IPC Seat Manage
  12 Lwe 60406818         124       2003      61 5300/6000   0 ARP Input
  13 Mwe 60581638           0          1       0 5760/6000   0 HC Counter Time
  14 Mwe 605E3D00           0          2       0 5564/6000   0 DDR Timers
  15 Msp 80164A38           0      79543       0 5608/6000   0 GraphIt
  16 Mwe 802DB0FC           0          2       0 11576/12000 0 Dialer event
  17 Cwe 801E74BC           0          1       0 5808/6000   0 Critical Bkgnd
  18 Mwe 80194D20           4       9549       0 10428/12000 0 Net Background
  19 Lwe 8011E9CC           0         20       0 11096/12000 0 Logger
  20 Mwe 80140160           8      79539       0 5108/6000   0 TTY Background
  21 Msp 80194114           0      95409       0 8680/9000   0 Per-Second Job
  22 Mwe 8047E960           0          2       0 5544/6000   0 dot1x
  23 Mwe 80222C8C           4          2    2000 5360/6000   0 DHCPD Receive
  24 Mwe 800844A0           0          1       0 5796/6000   0 HTTP Timer
  25 Mwe 80099378           0          1       0 5612/6000   0 RARP Input
  26 Mst 8022F178           0          1       0 11796/12000 0 TCP Timer
  27 Lwe 802344C8           0          1       0 11804/12000 0 TCP Protocols
  28 Hwe 802870E8           0          1       0 5784/6000   0 Socket Timers
  29 Mwe 80426048          64          3   21333 4488/6000   0 L2MM
  30 Mwe 80420010           4          1    4000 5592/6000   0 MRD
  30 Mwe 80420010           4          1    4000 5592/6000   0 MRD
  31 Mwe 8041E570           0          1       0 5584/6000   0 IGMPSN
  32 Hwe 80429B40           0          1       0 2604/3000   0 IGMP Snooping P
  33 Mwe 804F43B0           0          5       0 5472/6000   0 Cluster L2
  34 Mwe 804F18D0           0         17       0 5520/6000   0 Cluster RARP
  35 Mwe 804EA650           0         23       0 5440/6000   0 Cluster Base
  36 Lwe 802A1158           4          1    4000 5592/6000   0 Router Autoconf
  37 Mwe 80022058           0          1       0 5624/6000   0 Syslog Traps
  38 Mwe 8031CE88           0          1       0 5788/6000   0 AggMgr Process
  39 Mwe 8035EF88           0        407       0 5592/6000   0 PM Callback
  40 Mwe 80437B58           0          3       0 5556/6000   0 VTP Trap Proces
  41 Mwe 80027D40           0          2       0 5676/6000   0 DHCPD Timer
  42 Mwe 8040D3B0           0          2       0 2560/3000   0 STP STACK TOPOL
  43 Hwe 8040E338           0          2       0 2560/3000   0 STP FAST TRANSI
```

Task 3

• Create a simple network with two PCs connected via a router
. • Assign IP addresses to the PCs and router interfaces through the console.
• Pass messages (Ping) between the two PCs




```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

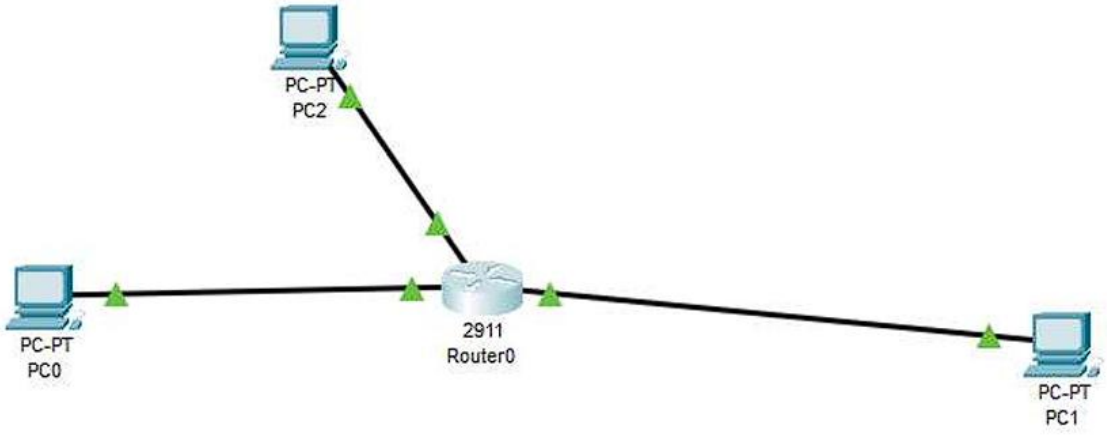
Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Task 4:

Add port to the router and connect pc to that port and check connection between the router and pc through ping command after giving ip addresses.






```
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.3: bytes=32 time<1ms TTL=127
Reply from 192.168.30.3: bytes=32 time<1ms TTL=127
Reply from 192.168.30.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Conclusion:

This lab successfully demonstrated foundational Cisco IOS device management and networking concepts by first utilizing show processes commands to observe the router's CPU and memory load, providing insights into its operational efficiency. Subsequently, a simple routed network was constructed, connecting two separate subnets via the router, verifying Layer 3 connectivity between the PCs using the ping command. Finally, the ability to dynamically add and configure a new router port to connect a third PC was proven, confirming connectivity between the new device and the existing network, thereby achieving the objectives Of resource inspection, inter-subnet communication, and port configuration.